

AN EXPLICIT OF(6, 4) AND AN EXPLICIT OF(8, 4)

ABSTRACT. An $\text{OF}(n, g)$ is a one-factorization of the complete multipartite graph $K_{n \times g}$ in which no one-factor contains two edges joining the same pair of parts; equivalently, an optimal decomposition of $K_{n \times g}$ with respect to distance 3. Tan, Zhou and Etzion state that for even $g \geq 4$ an $\text{OF}(n, g)$ exists if and only if $n > g$, *possibly except* odd n with $g + 3 \leq n \leq 2g - 3$ and even n with $g + 2 \leq n \leq 4g - 4$; at $g = 4$ that exception list is $\{6, 8, 10, 12\}$, and they settle $n = 10$. We exhibit a 12-edge base factor whose 20 translates under $\mathbb{Z}_5 \times \mathbb{Z}_4$ form an $\text{OF}(6, 4)$, and a 16-edge base factor whose 28 translates under $\mathbb{Z}_7 \times \mathbb{Z}_4$ form an $\text{OF}(8, 4)$. Both base factors are printed in full, and verifying each is a finite inspection of its twelve resp. sixteen edges together with three short lemmas. Granting the statements we quote from Tan, Zhou and Etzion, their doubling lemma turns the $\text{OF}(6, 4)$ into an $\text{OF}(12, 4)$ — which we neither reprove nor write out — and an $\text{OF}(n, 4)$ then exists for every $n > 4$. We do not determine whether an $\text{OF}(6, 4)$ or an $\text{OF}(8, 4)$ already follows from known amalgamation–detachment results.

1. DEFINITIONS AND THE QUOTED EXCEPTION LIST

For integers $n \geq 2$ and $g \geq 1$ let $K_{n \times g}$ be the complete multipartite graph with n parts of size g , so $|V(K_{n \times g})| = ng$ and $|E(K_{n \times g})| = g^2 \binom{n}{2}$. Two disjoint edges of $K_{n \times g}$ are *parallel* if their ends lie in the same pair of parts. Following Tan, Zhou and Etzion [2], an $\text{OF}(n, g)$ is a one-factorization of $K_{n \times g}$ in which no one-factor contains a pair of parallel edges; equivalently, an optimal decomposition of $K_{n \times g}$ with respect to distance 3, since a one-factor read as a set of words of length n and weight 2 over $\{0, 1, \dots, g\}$ has minimum Hamming distance ≥ 3 exactly when it has no two edges on one pair of parts. The notation carries $g > 1$ and $n > g$. The necessary conditions recorded in [2] are that ng be even and $n > g$; at $g = 4$ the parity clause is vacuous, so $n > 4$ is the whole of it.

Part (2) of the main existence theorem of [2] reads: for even $g \geq 4$ there exists an $\text{OF}(n, g)$ if and only if $n > g$, possibly except odd n with $g + 3 \leq n \leq 2g - 3$ and even n with $g + 2 \leq n \leq 4g - 4$. This clause and the other clauses of [2] used below are quoted from the e-print cited in the bibliography, not from a published version. At $g = 4$ the odd range $[g + 3, 2g - 3] = [7, 5]$ is empty and the even range $[g + 2, 4g - 4] = [6, 12]$ contributes $\{6, 8, 10, 12\}$, the four values that quoted statement leaves open at $g = 4$. Of these, $n = 10$ is settled inside [2] itself, by their $\text{OF}(g + 1, g)$ for even g (which gives an

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$\text{OF}(5, 4)$) followed by their doubling lemma. We exhibit objects for $n = 6$ and $n = 8$.

Theorem 1. *An $\text{OF}(6, 4)$ and an $\text{OF}(8, 4)$ exist. Both are exhibited in Section 2 as the orbit of a single one-factor under a group of order 20 resp. 28 acting freely on the edge set.*

Corollary 2. *Grant the clauses quoted from [2] above and the doubling lemma of [2] recorded as Lemma 7 below. Then an $\text{OF}(n, 4)$ exists if and only if $n > 4$.*

Only Theorem 1 is established here. The rest of Corollary 2 is quoted: $n = 12$ is Theorem 1 put through the doubling lemma of [2], no $\text{OF}(12, 4)$ being written out below, and every other $n > 4$ — including $n = 10$ — is already in [2].

We claim no priority for the two objects. We have not determined whether an amalgamation–detachment theorem controlling per-colour degrees and total multiplicities at once, such as that of Bahmanian and Rodger [1], yields these cells directly, and we have not established that no $\text{OF}(6, 4)$ or $\text{OF}(8, 4)$ appears in the literature. What is offered is two explicit objects with a verification a reader can carry out by hand.

2. THE TWO OBJECTS

Fix an odd integer $p \geq 3$ and an integer $g \geq 1$, and put $n = p + 1$. Take the parts of $K_{n \times g}$ to be indexed by $\mathbb{Z}_p \cup \{\infty\}$ and the g vertices of a part by $\mathbb{Z}_g$, so that

$$V = (\mathbb{Z}_p \cup \{\infty\}) \times \mathbb{Z}_g, \quad P_u = \{u\} \times \mathbb{Z}_g \quad (u \in \mathbb{Z}_p \cup \{\infty\})$$

and $E = E(K_{n \times g})$ is the set of pairs of vertices lying in different parts. Let

$$G = \mathbb{Z}_p \times \mathbb{Z}_g, \quad (t, s): (x, a) \mapsto (x + t, a + s), \quad \infty + t = \infty,$$

a group of order $pg = (n - 1)g$ acting on V , permuting the parts, and therefore acting on E .

An $\text{OF}(6, 4)$. Take $p = 5$, $g = 4$: six parts, 24 vertices, $|E| = \binom{24}{2} - 6\binom{4}{2} = 240$, and $|G| = 20$. Let M be the following one-factor of $K_{6 \times 4}$. Each line is one edge, written as **part value part value**, with **inf** for ∞ and values in $\mathbb{Z}_4$:

```

0 0  1 1
1 2  2 2
3 3  4 1
4 3  0 2
0 1  2 1
1 3  3 1
2 3  4 2
3 2  0 3
inf 0  1 0
inf 1  2 0
inf 2  3 0
inf 3  4 0

```

The 12 edges cover the 24 vertices once each, so M is a perfect matching, and each part contributes the four values 0, 1, 2, 3.

An OF(8, 4). Take $p = 7$, $g = 4$: eight parts, 32 vertices, $|E| = \binom{32}{2} - 8\binom{4}{2} = 496 - 48 = 448$, and $|G| = 28$. Let M_8 be

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0 0  1 0
0 1  2 1
0 2  4 1
0 3  5 0
1 1  2 2
1 2  inf 2
1 3  5 1
2 0  3 2
2 3  6 0
3 0  4 3
3 1  5 2
3 3  6 3
4 0  6 2
4 2  inf 1
5 3  inf 0
6 1  inf 3

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Again the 16 edges cover the 32 vertices once each and each part contributes 0, 1, 2, 3.

Theorem 3. *The 20 translates $M + \gamma$, $\gamma \in \mathbb{Z}_5 \times \mathbb{Z}_4$, form an OF(6, 4), and the 28 translates $M_8 + \gamma$, $\gamma \in \mathbb{Z}_7 \times \mathbb{Z}_4$, form an OF(8, 4).*

3. THE ARGUMENT

Theorem 3 follows from two lemmas that hold for every odd p and every g , plus a finite check on the printed edges.

Lemma 4. *G acts freely on E , and E splits into exactly $gn/2$ orbits, each of size $|G| = (n - 1)g$.*

Proof. Let $\gamma = (t, s) \in G$ fix an edge $e \in E$.

If $e = \{(x, a), (y, b)\}$ with $x, y \in \mathbb{Z}_p$, $x \neq y$, then either γ fixes each end or it interchanges them. In the second case $x + t = y$ and $y + t = x$, so $2t \equiv 0 \pmod{p}$; as p is odd, 2 is invertible modulo p , whence $t = 0$ and then $x = y$, a contradiction. In the first case $x + t = x$ gives $t = 0$ and $a + s = a$ gives $s = 0$, i.e. γ is the identity.

If $e = \{(\infty, a), (y, b)\}$ with $y \in \mathbb{Z}_p$, the two ends lie in different parts and ∞ is fixed by every element of G , so γ cannot interchange them; fixing them individually gives $t = 0$ and $s = 0$ as before.

So the stabiliser of every edge is trivial and every orbit has exactly $|G|$ elements. Consequently the number of orbits is

$$\frac{|E|}{|G|} = \frac{g^2 \binom{n}{2}}{(n-1)g} = \frac{gn}{2}. \quad \square$$

Since a one-factor of $K_{n \times g}$ has $ng/2$ edges, Lemma 4 says that the number of G -orbits on E equals the size of a one-factor; so a one-factor meeting every orbit at most once meets every orbit exactly once.

Lemma 5. *Let $M \subseteq E$ be a perfect matching of $K_{n \times g}$ such that*

- (T) *M contains exactly one edge from each G -orbit on E , and*
- (P) *the pairs of parts spanned by the edges of M are pairwise distinct.*

Then $\{M + \gamma : \gamma \in G\}$ is an OF(n, g).

Proof. Each element of G permutes V and maps parts onto parts, so every $M + \gamma$ is again a perfect matching of $K_{n \times g}$ and, by (P), again has pairwise distinct part-pairs: no $M + \gamma$ contains parallel edges.

The translates are pairwise edge-disjoint. Suppose $e_1 + \gamma_1 = e_2 + \gamma_2$ with $e_1, e_2 \in M$ and $\gamma_1, \gamma_2 \in G$. Then e_1 and e_2 lie in one orbit, so $e_1 = e_2$ by (T), and then $e_1 + \gamma_1 \gamma_2^{-1} = e_1$ forces $\gamma_1 = \gamma_2$ by Lemma 4. In particular the $|G|$ translates are pairwise distinct, and they carry

$$|G| \cdot \frac{ng}{2} = (n-1)g \cdot \frac{ng}{2} = g^2 \binom{n}{2} = |E|$$

distinct edges, i.e. they partition E . A partition of $E(K_{n \times g})$ into perfect matchings is a one-factorization, and none of its factors has parallel edges. $\square$

It remains to verify (T) and (P) for the two printed matchings, and for (T) we need to name the orbits. For $e \in E$ set

$$(1) \quad \iota(e) = \begin{cases} (d, b-a) & e = \{(x, a), (x+d, b)\}, \ x \in \mathbb{Z}_p, \ 1 \leq d \leq \frac{p-1}{2}, \\ (\infty, b-a) & e = \{(\infty, a), (y, b)\}, \ y \in \mathbb{Z}_p, \end{cases}$$

the first line being well defined because exactly one of the two orientations of a finite-finite edge has its part-difference in $\{1, \dots, (p-1)/2\}$.

Lemma 6. *ι is a complete invariant of the G -orbits on E : two edges lie in one orbit if and only if they have the same value of ι .*

Proof. ι is G -invariant, since (t, s) shifts the two part-indices of an edge by the same t and its two values by the same s . The fibre of ι over (d, j) with $1 \leq d \leq (p-1)/2$ is $\{(x, a), (x+d, a+j)\} : x \in \mathbb{Z}_p, a \in \mathbb{Z}_g\}$ and the fibre over (∞, j) is $\{(\infty, a), (y, a+j)\} : y \in \mathbb{Z}_p, a \in \mathbb{Z}_g\}$; each has exactly $pg = |G|$ elements. So ι has $g(\frac{p-1}{2} + 1) = \frac{g(p+1)}{2} = \frac{gn}{2}$ fibres, each of size $|G|$, and each fibre is a union of orbits. By Lemma 4 there are $gn/2$ orbits and each has size $|G|$, so each fibre is a single orbit. $\square$

Proof of Theorem 3. Both matchings were seen to be perfect matchings in Section 2, so by Lemmas 5 and 6 it suffices to read off ι and the part-pairs.

For M at $p = 5$ there are 12 orbits, indexed by $d \in \{1, 2\}$ and ∞ , each with four values of $b - a$ in $\mathbb{Z}_4$. In the order the edges are printed:

$$\begin{array}{llll} d = 1 : & (0,0)(1,1) & (1,2)(2,2) & (3,3)(4,1) & (4,3)(0,2) \\ & b-a=1 & 0 & 2 & 3 \\ d = 2 : & (0,1)(2,1) & (1,3)(3,1) & (2,3)(4,2) & (3,2)(0,3) \\ & b-a=0 & 2 & 3 & 1 \\ \infty : & (\infty,0)(1,0) & (\infty,1)(2,0) & (\infty,2)(3,0) & (\infty,3)(4,0) \\ & b-a=0 & 3 & 2 & 1 \end{array}$$

(the two entries in the fourth column of the first two rows use the orientations $4 \mapsto 0$ and $3 \mapsto 0$, of difference 1 and 2). Each of the three rows shows each element of $\mathbb{Z}_4$ exactly once, so all 12 values of ι are distinct and (T) holds. The part-pairs of M are

$$\{0,1\}, \{1,2\}, \{3,4\}, \{4,0\}, \{0,2\}, \{1,3\}, \\ \{2,4\}, \{3,0\}, \{\infty,1\}, \{\infty,2\}, \{\infty,3\}, \{\infty,4\},$$

twelve distinct pairs out of $\binom{6}{2} = 15$, so (P) holds.

For M_8 at $p = 7$ there are 16 orbits, indexed by $d \in \{1, 2, 3\}$ and ∞ . Reading (1) off the printed edges gives, grouped by index,

$$\begin{array}{l} d = 1 : (0,0)(1,0) \mapsto 0, (1,1)(2,2) \mapsto 1, (2,0)(3,2) \mapsto 2, (3,0)(4,3) \mapsto 3; \\ d = 2 : (0,1)(2,1) \mapsto 0, (3,1)(5,2) \mapsto 1, (4,0)(6,2) \mapsto 2, (0,3)(5,0) \mapsto 3; \\ d = 3 : (3,3)(6,3) \mapsto 0, (0,2)(4,1) \mapsto 1, (1,3)(5,1) \mapsto 2, (2,3)(6,0) \mapsto 3; \\ \infty : (1,2)(\infty,2) \mapsto 0, (4,2)(\infty,1) \mapsto 1, (6,1)(\infty,3) \mapsto 2, (5,3)(\infty,0) \mapsto 3. \end{array}$$

Four of the sixteen edges are read in the orientation that starts at the larger part label, that being the one whose difference lies in $\{1, 2, 3\}$: $(0,2)(4,1)$ from 4 to 0, since $4+3=0$ in $\mathbb{Z}_7$, giving $b-a=2-1=1$; $(0,3)(5,0)$ from 5 to 0, giving $3-0=3$; $(1,3)(5,1)$ from 5 to 1, giving $3-1=2$; and $(2,3)(6,0)$ from 6 to 2, giving $3-0=3$. Each of the four rows shows each element of $\mathbb{Z}_4$ exactly once, so (T) holds. The part-pairs of M_8 are the sixteen

$$\{0,1\}, \{0,2\}, \{0,4\}, \{0,5\}, \{1,2\}, \{1,5\}, \{1,\infty\}, \{2,3\}, \\ \{2,6\}, \{3,4\}, \{3,5\}, \{3,6\}, \{4,6\}, \{4,\infty\}, \{5,\infty\}, \{6,\infty\},$$

distinct, out of $\binom{8}{2} = 28$, so (P) holds. $\square$

4. WHAT FOLLOWS FROM THE QUOTED CLAUSES

Lemma 7 ([2], doubling lemma; quoted, not proved here). *Let n, g be positive integers with ng even and $n > g$. If there exists an $\text{OF}(n, g)$, then there exists an $\text{OF}(2n, g)$.*

At $(n, g) = (6, 4)$ we have $ng = 24$ even and $n = 6 > 4 = g$, which is all Lemma 7 as quoted asks, so Theorem 1 and that lemma give an $\text{OF}(12, 4)$; we do not write the object out. (The proof of Lemma 7 in [2] uses a pair of orthogonal Latin squares of order g and excludes $g = 2, 6$; both conditions are satisfied at $g = 4$. We have not reproved the lemma.)

Proof of Corollary 2. Necessity is the necessary condition of [2] quoted in Section 1: $n > g = 4$. For sufficiency, let $n > 4$. By part (2) of the existence theorem of [2], an $\text{OF}(n, 4)$ exists for every $n > 4$ outside the exception list $\{6, 8, 10, 12\}$. Within it, $n = 6$ and $n = 8$ are Theorem 1; $n = 12$ is Theorem 1 with Lemma 7; and $n = 10$ is the $\text{OF}(5, 4)$ of [2] with Lemma 7, whose hypotheses hold at $(n, g) = (5, 4)$ since $ng = 20$ is even and $5 > 4$. Every step but the two cases of Theorem 1 is a clause of [2] taken on trust. $\square$

5. VERIFICATION

Theorem 3 can be checked by hand from the two printed matchings: Lemmas 4–6 are three short arguments, and the two finite checks are the ι -tables and part-pair lists displayed in its proof.

The accompanying program `verify.py` also expands the factorizations. It parses the two blocks of Section 2 as printed, rebuilds $K_{6 \times 4}$, $K_{8 \times 4}$ and the two groups, verifies freeness, computes the orbits explicitly and confirms that ι is a complete invariant, and expands the 20 and 28 translates, checking that each is a perfect matching with pairwise distinct part-pairs and that the translates partition the 240 resp. 448 edges. Its recorded output (`verify.output.txt`) reports

VERDICT: ALL 56 CHECKS PASS

and states what it does not cover: it does not reprove Lemma 7 and constructs no $\text{OF}(12, 4)$; it takes the clauses of [2] used in Corollary 2 on trust, and checks their joint coverage only over the finite range $5 \leq n \leq 4000$; it does not check the wording quoted from [2] against any other version of that source; and it says nothing about minimality, uniqueness, priority or $g \geq 5$.

REFERENCES

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